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PROJECT TITLE

Energy Consumption Analytics Dashboard

Final Project Documentation

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CONTENTS

1. ABSTRACT 3
2. PROBLEM STATEMENT 4
3. OBJECTIVES 5
4. LITERATURE SURVEY 6
    4.1 Overview 6
    4.2 Literature Survey 7
5. RESEARCH GAP IDENTIFICATION 10
    5.1 Identified Research Gaps 10
    5.2 Link Between Research Gaps and Proposed Work 11
6. INNOVATION 12
7. CREATIVITY 13
8. NOVELTY 14
9. FEASIBILITY 15
10. PROPOSED FRAMEWORK 16
    10.1 FRAMEWORK DESCRIPTION 17
11. PROJECT PLAN 18
    11.1 Project Execution Phases 18
    11.2 Complete Workflow of Execution 19
    11.3 Expected Evaluation 20
12. CONCLUSION 21
13. REFERENCES 22

1. ABSTRACT

Commercial buildings combine heating, ventilation and air-conditioning (HVAC), lighting, plug loads, on-site photovoltaic generation, storage and grid imports. Their demand changes with weather, occupancy and electricity prices. This project documents a stateful building-energy digital twin that forecasts operating drivers, detects abnormal behavior, evaluates scenarios and recommends comfort-constrained control actions for a reference commercial office.

The reference site is a 7,500 m² single-zone office with a 250 kWp rooftop PV system, a 250 kWh battery and a reversible heat pump. The external drivers are real Berlin 2019 ERA5 weather, PVGIS-SARAH2 irradiance and PV output, and German/Luxembourg day-ahead electricity prices from the Open Power System Data snapshot. The building interior—including occupancy, lighting, plug loads, HVAC operation and indoor temperature—is simulated. No live building or real meter is connected.

The system ingests hourly data, maintains a timestamped twin state, identifies thermal parameters from telemetry, selects forecasts for outdoor temperature, irradiance and base load, scores injected faults, and evaluates non-mutating what-if scenarios. Pyomo and HiGHS solve a 24-hour comfort-constrained model predictive control problem. A local Streamlit dashboard presents system state and evaluation outputs.

In the repository's 78-day comparison, cost-focused MPC reduced operating cost from €5,868 to €5,440 (7.3%) and recorded zero comfort violations versus eight for the thermostat baseline. HVAC electricity rose from 21.81 to 26.24 MWh (+20.3%). The slide deck and decision log instead report +33.2% HVAC energy; the project records need reconciliation. Scenario analysis reports €428 baseline savings, €57 with no battery, and €720 when battery capacity is doubled. All outcomes are simulation results and should not be read as measured real-building savings.

2. PROBLEM STATEMENT

Commercial office energy management involves interacting systems. HVAC demand follows outdoor conditions and indoor comfort needs; lighting and plug loads follow building use; PV output varies with irradiance; battery dispatch changes grid imports; and electricity prices can vary hourly. When these subsystems are reviewed separately, operators have limited support for understanding how one action affects cost, energy use, peak demand and indoor conditions together.

A dashboard can display historical energy and weather, but it does not by itself represent the building's thermal state, estimate changing parameters, test abnormal behavior or compute constrained control actions. Forecasting, fault detection and optimization are often implemented as separate analyses, leaving the operator to combine their outputs manually. Scenario impacts are also difficult to assess without a repeatable simulation that preserves baseline inputs.

This project addresses the integration problem by creating a stateful digital representation of a commercial office. It receives timestamped telemetry, estimates thermal response, forecasts selected drivers, detects deviations, and schedules HVAC and battery operation subject to comfort and equipment constraints. The controller action advances the simulated plant, which then produces new telemetry for the twin.

The project is bounded to a single simulated building. Weather, satellite-derived irradiance/PV and wholesale prices are real external inputs; occupancy and building-interior measurements are simulated. Consequently, its purpose is to demonstrate an integrated method and support controlled experimentation—not to claim field deployment or guaranteed savings at a real site.

3. OBJECTIVES

The project objectives are to:

- Design a stateful digital twin for a commercial office containing HVAC, lighting, plug loads, PV, battery storage and grid interaction.
- Ingest hourly weather and price drivers together with simulated building telemetry, and maintain a timestamped state with data-quality information.
- Estimate building thermal parameters from observed telemetry while keeping the digital-twin code independent of simulator internals.
- Develop forecasting, residual-based anomaly detection and a flatline detector for sensor faults.
- Evaluate isolated what-if scenarios for battery, PV, occupancy, HVAC and comfort assumptions.
- Optimize HVAC and battery operation with comfort constraints in a 24-hour receding-horizon MPC loop.
- Compare thermostat and MPC operation using cost, HVAC energy, peak grid import and comfort violations.
- Present state, forecasts, anomaly information, scenarios and optimization outputs in a local Streamlit dashboard.

The scope is one simulated office and an offline/local prototype. A live building connection, physical sensing, multi-building service, and field deployment are outside the current implementation.

4. LITERATURE SURVEY

4.1 Overview

Building energy research spans data-driven forecasting, thermal modeling, fault detection and diagnostics, digital twins, and predictive control. Reviews of building-energy prediction describe machine-learning and statistical approaches for different end uses, time scales and data volumes [1], [2]. Their results are shaped by data coverage, occupancy representation, building type and the evaluation split.

Model predictive control (MPC) uses a model and forecasts to choose actions over a future horizon while respecting operating constraints. Building-focused reviews discuss model selection, comfort constraints, forecast quality, optimization choices and deployment challenges [3], [4]. Thermal parameter identification is important because a model that poorly represents the building may produce ineffective or uncomfortable schedules [5].

Automated fault detection and diagnostics (FDD) includes model-based, rule-based and process-history methods. Building-system reviews explain that anomaly flags need context and validation against known conditions [6], [7]. Isolation Forest is a general-purpose anomaly detector [8], but a building application still needs domain-specific interpretation and fault-specific scoring.

Digital-twin research emphasizes state synchronization and data flow between a physical asset and its digital representation [10], [11]. This project applies the closed-loop pattern in a simulation: telemetry updates the twin and control actions return to the plant model. The system is structurally bidirectional but does not represent a real connected building.

The literature review therefore focuses on integration. Forecasting, thermal identification, fault evaluation, scenario analysis and MPC are connected in one stateful workflow; their effects are measured together under simulation. The work does not claim that any individual model or optimization algorithm is new or universally superior.

4.2 Literature Survey

S.No.	Research Paper	Model / Approach	Dataset / Data Used + Link	Key Outcome	Limitation
1	Amasyali & El-Gohary (2018). "A review of data-driven building energy consumption prediction studies." Renewable and Sustainable Energy Reviews, 81, 1192–1205.	Review of data-driven prediction approaches and input variables.	Building-energy literature; multiple prediction tasks.	Summarizes drivers and learning methods used in building energy prediction.	Prediction review does not connect its forecasts to a closed-loop control system.
2	Ahmad et al. (2018). "A review of data-driven approaches for prediction and classification of building energy consumption." Renewable and Sustainable Energy Reviews, 82, 1027–1047.	Review of regression, machine learning and classification methods.	Building energy prediction and classification studies.	Organizes methods by application and building-analysis task.	Studies use varied datasets, targets and validation protocols.
3	Killian & Kozek (2016). "Ten questions concerning model predictive control for energy efficient buildings." Building and Environment, 105, 403–412.	Critical discussion of building MPC design and application.	Prior building MPC studies and deployments.	Highlights model quality, comfort, forecasts and implementation considerations.	MPC performance depends on models, forecasts and practical integration.
4	Drgoňa et al. (2020). "All you need to know about model predictive control for buildings." Annual Reviews in Control, 50, 190–232.	Comprehensive review of building MPC formulations.	Building climate control literature.	Maps modeling, optimization and control approaches.	Real deployment must handle data, model and integration constraints.

4.2 Literature Survey (Continued)

S.No.	Research Paper	Model / Approach	Dataset / Data Used + Link	Key Outcome	Limitation
5	Bacher & Madsen (2011). "Identifying suitable models for the heat dynamics of buildings." Energy and Buildings, 43(7), 1511–1522.	Comparison of low-order models for thermal response.	Building heat-dynamics data.	Shows the value of selecting model structure for building behavior.	Parameter estimates depend on excitation and quality of measured signals.
6	Katipamula & Brambley (2005), Part I. "Methods for fault detection, diagnostics, and prognostics for building systems." HVAC&R Research, 11(1), 3–25.	Review and taxonomy of FDD methods.	Building-system and general engineered-system research.	Classifies methods and discusses method strengths and weaknesses.	General method categories do not guarantee fault-specific sensitivity.
7	Katipamula & Brambley (2005), Part II. "Methods for fault detection, diagnostics, and prognostics for building systems." HVAC&R Research, 11(2), 169–187.	Review focused on HVAC&R applications.	HVAC&R fault-detection research.	Reviews system-specific methods and deployment state.	Operational adoption and independent validation remain challenging.

4.2 Literature Survey (Continued)

S.No.	Research Paper	Model / Approach	Dataset / Data Used + Link	Key Outcome	Limitation
8	Liu, Ting & Zhou (2008). "Isolation Forest." IEEE International Conference on Data Mining, 413–422.	Unsupervised tree-based anomaly detection.	Generic multivariate anomaly-detection problem.	Uses random partitioning to isolate anomalous observations.	A generic score does not identify the physical or sensor cause.
9	Oldewurtel et al. (2012). "Use of model predictive control and weather forecasts for energy efficient building climate control." Energy and Buildings, 45, 15–27.	Forecast-aware MPC for indoor climate control.	Building climate control with weather predictions.	Studies energy use and comfort under predictive operation.	Findings depend on specific building and model assumptions.
10	Fuller et al. (2020). "Digital twin: Enabling technologies, challenges and open research." IEEE Access, 8, 108952–108971.	Cross-domain digital-twin review.	Digital-twin concepts and enabling technologies.	Summarizes twin architecture, synchronization and open issues.	Broad coverage does not provide a building-specific energy controller.

5. RESEARCH GAP IDENTIFICATION

5.1 Identified Research Gaps

The reviewed literature provides methods for individual analytics and control tasks. The project identifies the following integration and evaluation gaps within its scope:

S.No.	Research Gap	Why It Matters	Project Response
1	Disconnected analytics	Forecasting, fault detection, scenario analysis and control are often evaluated in separate workflows.	Connect each capability through a stateful telemetry loop.
2	Twin validation boundary	A simulation can overstate independence if the twin reads plant internals.	The twin receives telemetry only; code-level separation is enforced.
3	Cost-energy trade-off	Cost-optimal operation can increase HVAC electricity and peak demand.	Report cost, energy, peak and comfort together; compare objective variants.
4	Battery contribution	Storage value depends on capacity, tariff and operating constraints.	Measure battery removal and capacity increases in scenarios.
5	Stuck sensor blindness	Residual methods may miss a sensor that freezes prediction inputs.	Add a flatline detector and score known injected faults.
6	Real-world evidence	Interior building signals and the plant are simulated.	Treat findings as simulation outcomes; validate later on a real site.

5.2 Link Between Research Gaps and Proposed Work

The proposed system responds to the integration gap by connecting exogenous data, simulated plant telemetry, twin-state updates, forecasting, fault scoring, scenario analysis and optimization. The controller returns its first action to the simulated plant, which advances and produces subsequent telemetry. This demonstrates a closed computational loop while preserving a strict information boundary between the twin and the simulator.

The project evaluates the consequences of cost-focused operation across multiple metrics. In the repository's 78-day table, cost falls by 7.3% while HVAC electricity increases by 20.3%, peak demand rises, and comfort violations fall from eight to zero. A cost-plus-energy objective and a cost-plus-peak objective provide distinct alternatives. The presentation's +33.2% energy figure conflicts with the repository's +20.3% value and remains to be reconciled.

Scenario results quantify the role of system configuration. The no-battery case reduces savings to €57 (0.98%); doubling the battery raises savings to €720 (12.27%). A wider ±1 K comfort band reports €469 savings and reduces HVAC energy to 20.52 MWh. These outputs are conditional on the reference building, tariff, test period and simulator assumptions.

The fault experiments illustrate why validation should be fault-specific. Residual-based methods recorded zero recall on the injected stuck-temperature-sensor case; a flatline detector reached 0.944 recall with four-hour latency. This finding motivates complementary detection signals and additional independent measurements for diagnosis. It does not establish the same recall for field faults.

The project therefore contributes an integrated simulation and an explicit record of its trade-offs and limits. Future work must connect measured site data, verify safety and comfort under uncertainty, and reconcile the two reported HVAC energy changes before external claims are made.

6. INNOVATION

The proposed innovation is an integrated and measurable energy-management loop for a simulated office. The work combines established techniques in a coordinated system rather than claiming a new forecasting or optimization algorithm.

Stateful Building Twin

Maintains a timestamped view of the building, zones, assets and sensors, including quality and staleness information.

Telemetry-Only Model Boundary

The twin updates from noisy and occasionally missing observations; it does not inspect simulator state.

Joint Forecast, Fault and Control Workflow

Driver forecasts and residual analysis inform scenario evaluation and comfort-constrained MPC.

Multi-Objective Reporting

Cost, HVAC electricity, peak import and comfort violations are shown together so a cost reduction is not mistaken for an energy reduction.

Measured Scenario Sensitivity

Battery capacity, PV, occupancy, HVAC capacity and comfort bounds can be varied without mutating baseline inputs.

Fault-Specific Detector Evaluation

Known injected faults support recall and latency measurement; flatline detection covers a stuck-sensor case that residual methods miss.

7. CREATIVITY

The project presents energy operation as an interactive scenario rather than a static dashboard. The local interface ties together data, asset state, energy flows and control comparisons so a user can move from “what is happening?” to “what changes if the operating plan changes?”

Building State View

Shows current and historical values for indoor/outdoor conditions, demand, PV, grid import, assets and telemetry health.

Forecast and Asset Views

Compares measured and forecast drivers and exposes model calibration and asset metadata.

Scenario Lab

Lets the user explore battery, PV, comfort, occupancy and HVAC capacity changes while preserving baseline scenario inputs.

Control Comparison

Compares thermostat, greedy battery and MPC variants across cost, HVAC energy, peak demand and comfort.

Anomaly Review

Displays detected deviations alongside fault context and evaluation information.

Closed-Loop Demonstration

Applies the first MPC action to the simulated plant; resulting telemetry flows back into the twin.

The prototype uses a Streamlit interface with Plotly for interactive views and matplotlib for static project figures. FastAPI, MQTT, Docker and database services are deferred; the interface is a local demonstration rather than a deployed building-management system.

8. NOVELTY

Digital-twin architectures, predictive control, load forecasting and fault detection are established research areas. The project novelty lies in applying these methods together to a documented reference office and making the system's evidence boundary explicit.

Bidirectional Simulation Loop

Telemetry updates the twin, and the selected control action advances the simulated plant for the next update.

Twin and Plant Separation

The digital twin is architecturally independent from simulator internals, allowing it to operate on a telemetry interface.

Trade-Off Transparency

The evaluation reports cost alongside HVAC energy, peak demand and comfort; cost savings alone are not treated as energy savings.

Storage Sensitivity

Scenario runs test whether battery or building thermal mass contributes more flexibility under the chosen tariff and comfort constraints.

Residual and Flatline Complementarity

The project shows how a frozen sensor can evade residual checks and why an absence-of-variation detector is useful.

Evidence-Limited Claims

The report identifies the building as simulated and records conflicts in the energy-percentage values rather than presenting them as resolved.

9. FEASIBILITY

The project is feasible as a software prototype because its data, simulation, analysis and interface components run in a local Python environment. It does not require physical sensors or building access for the current scope.

Technical

Python 3.11+, pandas, NumPy, scikit-learn, Pyomo, HiGHS, Plotly and Streamlit support the implemented pipeline.

Data

The committed driver series provides 8,758 hourly rows for 2019. Weather, irradiance/PV and wholesale price inputs are sourced; indoor loads and building states are simulated.

Computational

The project uses an hourly single-zone model and a 24-hour MPC horizon. The model and scenarios can be run on a standard development computer; no physical hardware is required.

Implementation

The archive includes data ingestion, a simulated plant, twin state, thermal calibration, forecasts, anomaly scoring, scenarios, optimization and dashboard modules.

Economic

The prototype uses open-source code and public data services. Data attribution and reuse conditions should be checked before redistribution.

Schedule

Work can be organized as data verification, plant/twin implementation, analytics, optimization, dashboard integration and evaluation.

Operational

The current interface is local and demonstrative. Secure APIs, authentication, live data acquisition, alarms and operator approval are required before deployment.

10. PROPOSED FRAMEWORK

The framework combines real external drivers with a simulated office interior. Telemetry enters a stateful twin and is used for thermal identification, driver forecasting and fault analysis. A scenario engine and comfort-constrained MPC generate candidate operating actions, which advance the simulated plant and produce the next observations.

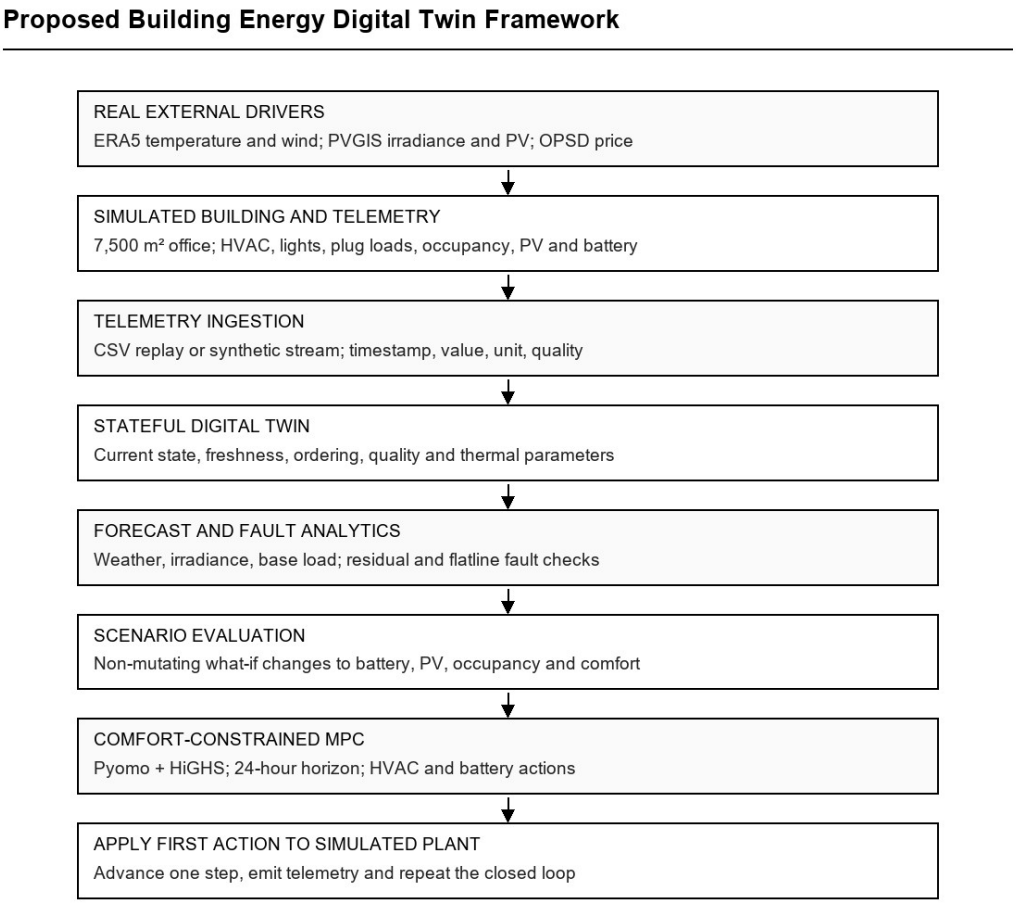


Fig. 1. Proposed framework for a stateful building energy digital twin.

10.1 FRAMEWORK DESCRIPTION

The project driver dataset contains 8,758 hourly rows from January 1 through December 31, 2019 in UTC. PVGIS irradiance timestamps are floored to the hour before joining with the OPSD DE/LU day-ahead price series [12]–[14]. Two hours without a published price are omitted. Outdoor weather and PV inputs are real external data; indoor temperature, occupancy, loads and HVAC behavior are simulated. Driver forecasting uses a chronological split of 5,832 training hours, 1,056 validation hours and 1,870 test hours; the model-selection results cover 1,032 evaluation samples.

Framework Component	Description / Project Setting
1. Data sources	Load PVGIS/ERA5 weather and PV series and OPSD day-ahead prices.
2. Telemetry	Replay CSV drivers or use a synthetic stream; attach timestamps, values, units and quality.
3. Twin state	Maintain valid/latest observations, ordering and staleness across building entities.
4. Thermal model	Estimate UA and heat capacity from telemetry without using hidden plant state.
5. Forecasts	Select Ridge for outdoor temperature and base load, and seasonal mean for irradiance.
6. Fault detection	Score residuals, rolling statistics and flatline behavior on injected faults.
7. Scenarios	Run isolated changes in occupancy, PV, battery, HVAC, price and comfort assumptions.
8. MPC	Solve a 24-hour horizon; apply the first HVAC/battery control action and repeat.

The thermal model identified UA = 5.83 kW/K and capacitance = 357 kWh/K, while the simulated plant uses 5.00 kW/K and 250 kWh/K. This is +17% and +43% respectively. The project reports an identified time constant of 61 h versus 50 h in the plant. The twin's model bias and its effect on comfort are included in the evaluation, not concealed.

11. PROJECT PLAN

The work proceeds from source and data verification through the digital-twin loop, analytics, optimization, interface integration and scenario-based evaluation.

11.1 Project Execution Phases

Phase	Work to be Completed	Lead
Phase 1	Confirm goals, reference office, equipment sizes, comfort scope and simulated/real-data boundary.	Team
Phase 2	Verify PVGIS/ERA5 weather/PV and OPSD price provenance; align hourly UTC data.	B. Vignesh Reddy
Phase 3	Build the simulated office and quality-tagged telemetry/replay interface.	B. Vignesh Reddy
Phase 4	Implement twin entities, state updates, thermal identification and calibration.	P. Karthik
Phase 5	Evaluate driver forecasts, injected-fault detection and scenario isolation.	K. Divya Charan
Phase 6	Formulate comfort-constrained HVAC/battery MPC and objective variants.	Team integration
Phase 7	Integrate local Streamlit dashboard and interactive plots.	Team integration
Phase 8	Compare baseline/controller outcomes, record limitations and prepare documentation.	Team

11.2 Complete Workflow of Execution

The workflow keeps driver preparation, twin updates, analysis and control in chronological order. At each hourly step the controller acts on the simulated building; the next telemetry record becomes the new twin observation.

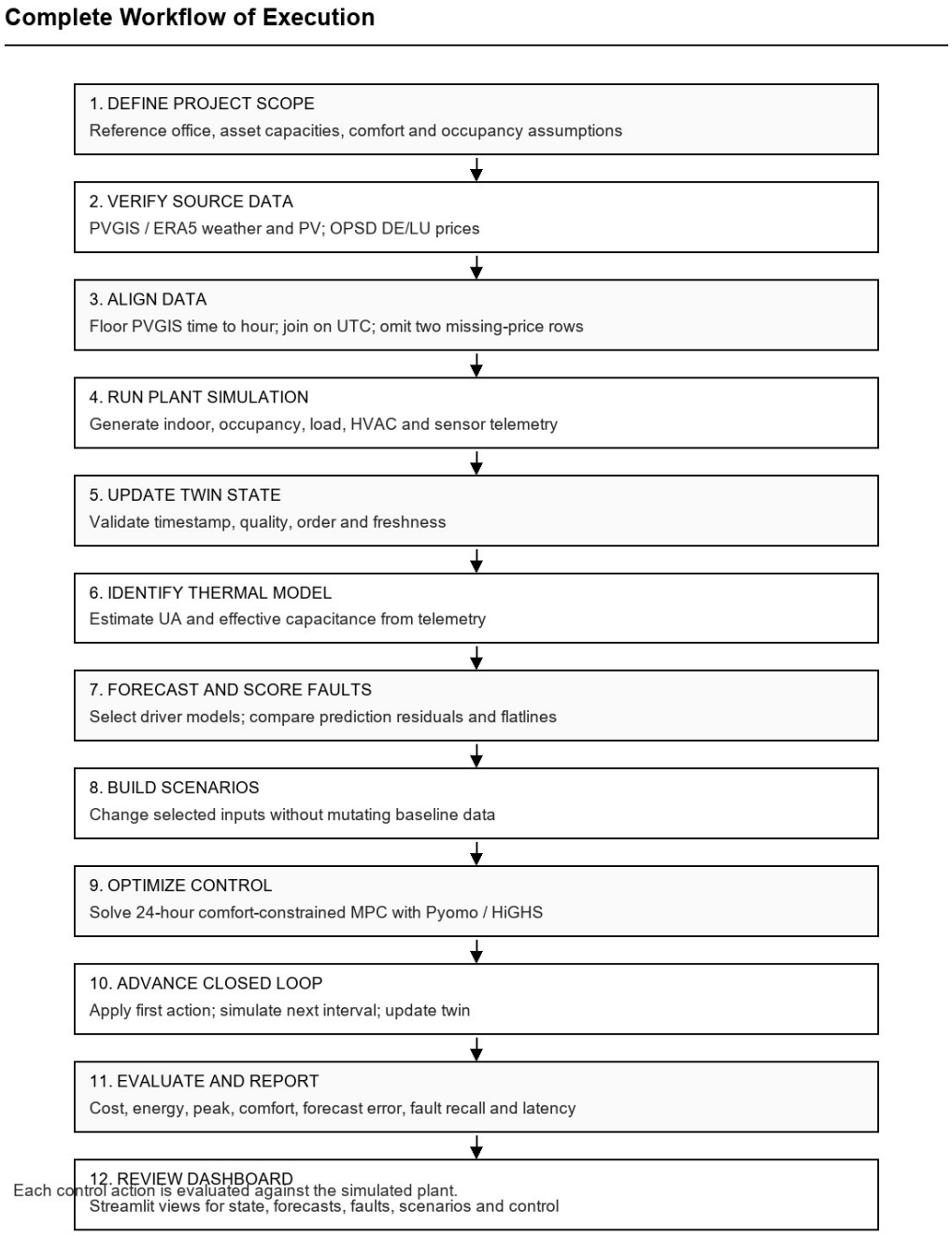


Fig. 2. Complete project execution workflow from scope definition to dashboard review.

11.3 Expected Evaluation

The system should be evaluated across the following criteria:

- Forecast MAE, RMSE, R^2 and skill relative to persistence on a chronological evaluation set.
- Fault precision, recall, F1, specificity, false-positive rate and detection latency for known injected fault types.
- Operating cost, HVAC electricity, peak grid import and comfort violations for thermostat and MPC variants.
- Scenario sensitivity to battery capacity, PV, occupancy, HVAC efficiency/capacity, tariff and comfort-band changes.
- Telemetry handling, state freshness, scenario input isolation and closed-loop data/control flow.
- Dashboard usefulness, clear source attribution, explanation of uncertainty and suitability for operator review.

Reported Simulation Results

Metric	Thermostat	MPC Cost	Interpretation
Cost over 78-day window	€5,868	€5,440	€428 lower; 7.3% saving
HVAC electricity	21.81 MWh	26.24 MWh	+20.3% in repository result table
Peak grid import	337 kW	400 kW	MPC cost-only variant has higher peak
Comfort violations	8	0	No violation recorded for this MPC run

The project slide deck and decision log report a +33.2% HVAC-energy increase and a −27.1% HVAC-cost index. The repository's 78-day result table instead supports +20.3% HVAC energy and a 7.3% total operating-cost reduction. These figures may come from different runs or definitions; the source records should be reconciled before formal submission or publication. Scenario-sweep savings (baseline €428, no battery €57, doubled battery €720) are reported separately and should not be mixed with other test-window indices.

The stuck-temperature-sensor detector achieved 0.944 recall with four-hour latency; residual-based detectors scored zero recall on this fault. The project also reports that true-parameter MPC saved 7.9% but caused 269 comfort violations, compared with 7.3% savings and zero violations using the identified model. These are simulation-specific outcomes. Full test-suite totals are not asserted here because the presentation states that a total pass count is unavailable.

12. CONCLUSION

This project documents a stateful building-energy digital twin for a simulated 7,500 m² office. The system links hourly weather and price drivers, simulated plant telemetry, thermal-state estimation, forecasting, fault detection, isolated scenarios and comfort-constrained MPC. A local Streamlit dashboard exposes state and evaluation outputs.

The 78-day repository comparison reports €428 (7.3%) lower operating cost under cost-focused MPC, alongside zero comfort violations instead of eight for the thermostat. HVAC electricity increases from 21.81 to 26.24 MWh (+20.3%) in the result table, illustrating why energy and cost must be reported together. Scenario analysis shows strong sensitivity to battery size, and a flatline detector handles a stuck-sensor case missed by residual methods.

The plant and building interior are simulated, so the results do not establish real-site savings or field readiness. The presentation's +33.2% energy increase conflicts with the repository table and must be reconciled. Future work should connect measured building telemetry, use multi-zone and uncertainty-aware models, evaluate longer periods and realistic tariffs, expand fault diagnosis, and add secure operator-approved deployment controls.

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